

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Third Semester B.Tech (Biotechnology) (CBCS) Degree Examinations

December 2023

BT1001-1 – BIOCHEMISTRY

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Which of the following is a polymer of chitin?
 A) N-acetyl galactosamine
 B) beta-D galactopyranosyl
 C) N-acetyl glucosamine
 D) beta-glucose
- More complex carbohydrate polymers covalently attached to proteins or lipids, is called
 A) Glycoconjugates
 B) Proteoglycans
 C) Chitin
 D) Sulfolipids
- _____ is a polysaccharide forming the exoskeletons of many invertebrates.
 A) Chitin
 B) Glucose
 C) Cellulose
 D) Glycogen
- Amylopectin and cellulose are compared and both these polysaccharide _____
 A) function as structural components
 B) contain α - glycosidic bond
 C) have extended linear shapes
 D) contain 1,4-glycosidic bond
- A steroid molecule consists of how many fused carbon rings?
 A) 2
 B) 4
 C) 3
 D) 5
- The number of milligrams of KOH required to neutralize the free fatty acids of 1 gram of fat is called
 A) Acid number
 B) Iodine number
 C) Saponification number
 D) Saturation
- Progesterones are
 A) Lipids
 B) Carbohydrates
 C) Amino acids
 D) Proteins
- Structural lipid of cell membrane is
 A) Simple lipid
 B) Steroid
 C) Vitamins derivatives
 D) Phospholipid
- Peptide bond is a _____
 A) Ionic bond
 B) Metallic bond
 C) Covalent bond
 D) Hydrogen bond
- Which of the following is not a conjugated protein?
 A) Complete proteins
 B) Lipoproteins
 C) Glycoproteins
 D) Metalloproteins
- Name the amino acid, which exists in two non-superimposable mirror images of each other.
 A) Epimer
 B) Anomer
- What is the maximum wavelength that Tryptophan and tyrosine absorb?
 A) 280nm
 B) 260nm
 C) 257nm
 D) 230nm
- Which of the following purine bases is present in RNA?
 A) Uracil
 B) Guanine
 C) Thymine
 D) Cytosine

P.T.O.

14. Which of the following RNAs are the most abundant in an animal cell?
 A) rRNA B) mRNA
 C) tRNA D) miRNA
15. Which of the following RNAs' structure is similar to clover leaf?
 A) rRNA B) mRNA
 C) tRNA D) hnRNA
16. The bases are held together in a DNA double helix by hydrogen bonds. These bonds are
 A) Ionic bonds B) Covalent bonds
 C) Van der Waals forces D) Non-covalent bonds
17. Number of ATP formed at the end of glycolysis during substrate level phosphorylation
 A) 28 B) 16
 C) 2 D) 42
18. Which of the following biomolecules is common to respiration mediated breakdown of fats, carbohydrates, and protein
 A) Pyruvic acid B) Acetyl CoA
 C) Fructose 1,6- biphosphate D) Glucose-6-phosphate
19. Kreb's cycle is also known as
 A) TCA Cycle B) Glycolysis
 C) Pentose phosphate pathway D) Gluconeogenesis
20. Glycolysis takes place in _____ and the end product is _____.
 A) Mitochondria, 2 Pyruvate B) Cytoplasm, 2 Pyruvate
 C) Cytoplasm, 3 Pyruvate D) Hepatocyte, 2 Pyruvate

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Briefly discuss monosaccharides and comment on glucose enantiomers.	8	L*2	1	1
b)	Give an account on different types of sterols and its role in biological system.	8	L2	2	1
2. a)	Explain the role of polysaccharides in invertebrate exoskeletons and plant cell walls. Depict its structure.	8	L2	1	1
b)	What are triglycerides? Triacylglycerols provide stored energy and insulation. Discuss.	8	L2	2	1
3. a)	Give an account on structural properties of starch and glycogen.	8	L2	1	1
b)	Write a note on general properties of lipids.	8	L2	2	1
Unit – II					
4. a)	Differentiate fibrous and globular proteins.	8	L2	3	1
b)	Give an account on Pseudoknot structure in RNA.	8	L2	4	1
5. a)	Discuss forces that determine protein structures.	8	L2	3	1
b)	Explain "cloverleaf" model of tRNA.	8	L2	4	1
6. a)	Depict peptide bond formation between two amino acids. Give an account on classification of amino acids based on polarity of side chains.	8	L2	3	1
b)	Elucidate structure of nucleosomes.	8	L2	4	1
Unit – III					
7. a)	Discuss salvage pathway.	8	L2	5	1
b)	Depict pathway of gluconeogenesis leading to synthesis of glucose. Mention its significance.	8	L2	5	1
8. a)	Pentose sugars play a major role in cellular metabolism. Depict pentose phosphate pathway and list its advantages.	8	L2	5	1
b)	How is urea formed in the cellular system? Explain the pathway.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
